

Healthcare Analytics Dashboard Project Report

This report explains the complete Healthcare Analytics Dashboard project created using Excel/LibreOffice. The project includes data cleaning, transformation, pivot tables, dashboard creation, and healthcare insights analysis.

Project Objectives

- Clean and preprocess healthcare-related data.
- Handle missing and inconsistent values.
- Transform raw data into analytical insights.
- Create pivot tables and charts.
- Build an interactive healthcare dashboard.

Dataset Sheets

- Customer Names
- Medical Examinations
- Hospitalisation Details
- Pivot Tables
- Dashboard

Data Cleaning & Transformation

- Handled missing values and incorrect entries.
- Created BMI and Diabetes Status categories.
- Merged multiple sheets using lookup functions.
- Calculated Age using Date of Birth.
- Converted healthcare charges into currency format.

Dashboard KPIs

- Total Customers
- Average Charges
- Average BMI
- Average HbA1C
- Total Smokers

Customer Names Data Transformation

	A	B	C	D	E	F
1	Customer ID	Title	First Name	Last Name		
2	Id1	Mr.	Kelly	Hawks		
3	Id2	Mr.	Matthew D	Lehner		
4	Id3	Mr.	Phil	Lu		
5	Id4	Ms.	Kelsey	Osborne		
6	Id5	Ms.	Russell B.	Kadala		
7	Id6	Mr.	Russell B.	Baker		
8	Id7	Mr.	Scott	Macpherson		
9	Id8	Mr.	Stephen	Hallman		
10	Id9	Mr.	Patrick R.	Moran		
11	Id10	Ms.	Brooke N.	Benner		
12	Id11	Ms.	Paola Andrea	Fierro Vargas		
13	Id12	Mr.	David	Franz		
14	Id13	Mr.	Wade	Foster		
15	Id14	Mr.	Franklin	Tenorio		
16	Id15	Ms.	Leilani M.	Rios		
17	Id16	Mr.	Philippe	Viau-Dupuis		
18	Id17	Ms.	Jennifer A.	Cronin		
19	Id18	Ms.	Christina M.	Noordstar		
20	Id19	Mr.	Said Sr.	Boudalia		
21	Id20	Mr.	John	Flor		
22	Id21	Mr.	Myles	Fennon		
23	Id22	Ms.	Madelyn C	Hribar		
24	Id23	Ms.	Nicole	Tassello		
25	Id24	Mr.	Eric A.	Mauricette		
26	Id25	Mr.	Emiliano J.	Garcia		

Medical Examination Dataset

	A	B	C	D	E	F	G	H	I	J
1	Customer	BMI	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	Weight status	Diabetes Status	smoker
2	Id1	47.41	7.47	No	No	No		0 Obesity	Diabetes	yes
3	Id2	30.36	5.77	No	No	No		0 Obesity	Diabetes	yes
4	Id3	34.485	11.87	yes	No	No		2 Obesity	Prediabetes	yes
5	Id4	38.095	6.05	No	No	No		0 Obesity	Diabetes	yes
6	Id5	35.53	5.45	No	No	No		0 Obesity	Normal	yes
7	Id6	32.8	6.59	No	No	No		0 Obesity	Prediabetes	yes
8	Id7	36.4	6.07	No	No	No		0 Obesity	Diabetes	yes
9	Id8	36.96	7.93	No	No	No		3 Obesity	Prediabetes	yes
10	Id9	41.14	9.58	yes	No	Yes		1 Obesity	Diabetes	yes
11	Id10	38.06	10.79	No	No	No		0 Obesity	Diabetes	yes
12	Id11	37.7	5.96	yes	No	No		2 Obesity	Diabetes	yes
13	Id12	42.13	11.9	No	No	No		0 Obesity	Prediabetes	yes
14	Id13	40.92	8.41	No	No	No		0 Obesity	Diabetes	yes
15	Id14	40.565	7.02	No	No	No		0 Obesity	Diabetes	yes
16	Id15	36.385	7.59	yes	No	No		2 Obesity	Diabetes	yes
17	Id16	39.9	11.32	No	No	No		0 Obesity	Diabetes	yes
18	Id17	33.8	7.67	No	No	No		3 Obesity	Diabetes	yes
19	Id18	36.765	7.29	yes	No	Yes		1 Obesity	Diabetes	yes
20	Id19	36.955	4.72	yes	No	No		1 Obesity	Normal	yes
21	Id20	42.9	11.41	No	No	No		0 Obesity	Prediabetes	yes
22	Id21	36.3	11.5	yes	No	No		2 Obesity	Diabetes	yes
23	Id22	32.2	6.22	yes	No	No		2 Obesity	Diabetes	yes
24	Id23	31.3	11.38	No	No	No		3 Obesity	Prediabetes	yes
25	Id24	41.8	7.89	yes	No	No		2 Obesity	Diabetes	yes
26	Id25	35.09	4.38	yes	No	No		2 Obesity	Normal	yes
27	Id26	33.88	7.01	No	No	No		3 Obesity	Prediabetes	yes
28	Id27	30.875	11.88	No	No	No		0 Obesity	Diabetes	yes
29	Id28	36.86	5.19	yes	No	Yes		1 Obesity	Normal	yes

Hospitalisation Details Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID	Date of Birth	Age		
2	ld2335	1992	Jul	9	0	₹563.84	tier - 2	tier - 3	R1013	30/11/1992	30	1982.84	
3	ld2334	1992	Nov	30	0	₹570.62	tier - 2	tier - 1	R1013	30/11/1992	30		
4	ld2333	1993	Jun	30	0	₹600.00	tier - 2	tier - 1	R1013	30/06/1993	29		
5	ld2332	1992	Sep	13	0	₹604.54	tier - 3	tier - 3	R1013	13/09/1992	30		
6	ld2331	1998	Jul	27	0	₹637.26	tier - 3	tier - 3	R1013	27/07/1998	24		
7	ld2330	2001	Nov	20	0	₹646.14	tier - 3	tier - 3	R1012	20/11/2001	21		
8	ld2329	1993	Jun	1	0	₹650.00	tier - 3	tier - 3	R1013	01/06/1993	30		
9	ld2328	1995	Jul	4	0	₹650.00	tier - 3	tier - 3	R1013	04/07/1995	27		
10	ld2327	2002	Nov	29	0	₹668.00	tier - 3	tier - 2	R1012	29/11/2002	20		
11	ld2326	1997	Nov	9	0	₹670.00	tier - 3	tier - 3	R1013	09/11/1997	25		
12	ld2325	2001	Sep	12	0	₹687.54	tier - 3	tier - 2	R1013	12/09/2001	21		
13	ld2324	1999	Dec	26	0	₹700.00	tier - 3	tier - 3	R1013	26/12/1999	23		
14	ld2323	1999	Dec	14	0	₹722.99	tier - 3	tier - 1	R1013	14/12/1999	23		
15	ld2322	2002	Sep	19	0	₹750.00	tier - 3	tier - 1	R1012	19/09/2002	20		
16	ld2321	1993	Aug	9	0	₹760.00	tier - 3	tier - 1	R1013	09/08/1993	29		
17	ld2320	1996	Oct	22	0	₹760.00	tier - 3	tier - 3	R1013	22/10/1996	26		
18	ld2319	1993	Jun	28	0	₹770.00	tier - 3	tier - 3	R1013	28/06/1993	29		
19	ld2318	1996	Sep	18	0	₹770.38	tier - 3	tier - 3	R1012	18/09/1996	26		
20	ld2317	1995	Dec	7	0	₹773.54	tier - 3	tier - 2	R1013	07/12/1995	27		
21	ld2316	2004	Oct	7	0	₹830.52	tier - 3	tier - 2	R1011	07/10/2004	18		
22	ld2315	2000	Nov	18	0	₹865.41	tier - 3	tier - 1	R1013	18/11/2000	22		
23	ld2314	1993	Nov	27	0	₹896.21	tier - 3	tier - 1	R1013	27/11/1993	29		
24	ld2313	1994	Oct	30	0	₹915.07	tier - 3	tier - 1	R1013	30/10/1994	28		
25	ld2312	1995	Oct	28	0	₹928.59	tier - 3	tier - 1	R1013	28/10/1995	27		
26	ld2311	2001	Aug	19	0	₹964.71	tier - 3	tier - 2	R1013	19/08/2001	21		
27	ld2310	1994	Nov	22	0	₹1,006.65	tier - 3	tier - 2	R1013	22/11/1994	28		
28	ld2309	2000	Oct	9	0	₹1,012.00	tier - 3	tier - 2	R1013	09/10/2000	22		
29	ld2308	1999	Aug	5	0	₹1,019.00	tier - 3	tier - 2	R1013	05/08/1999	23		

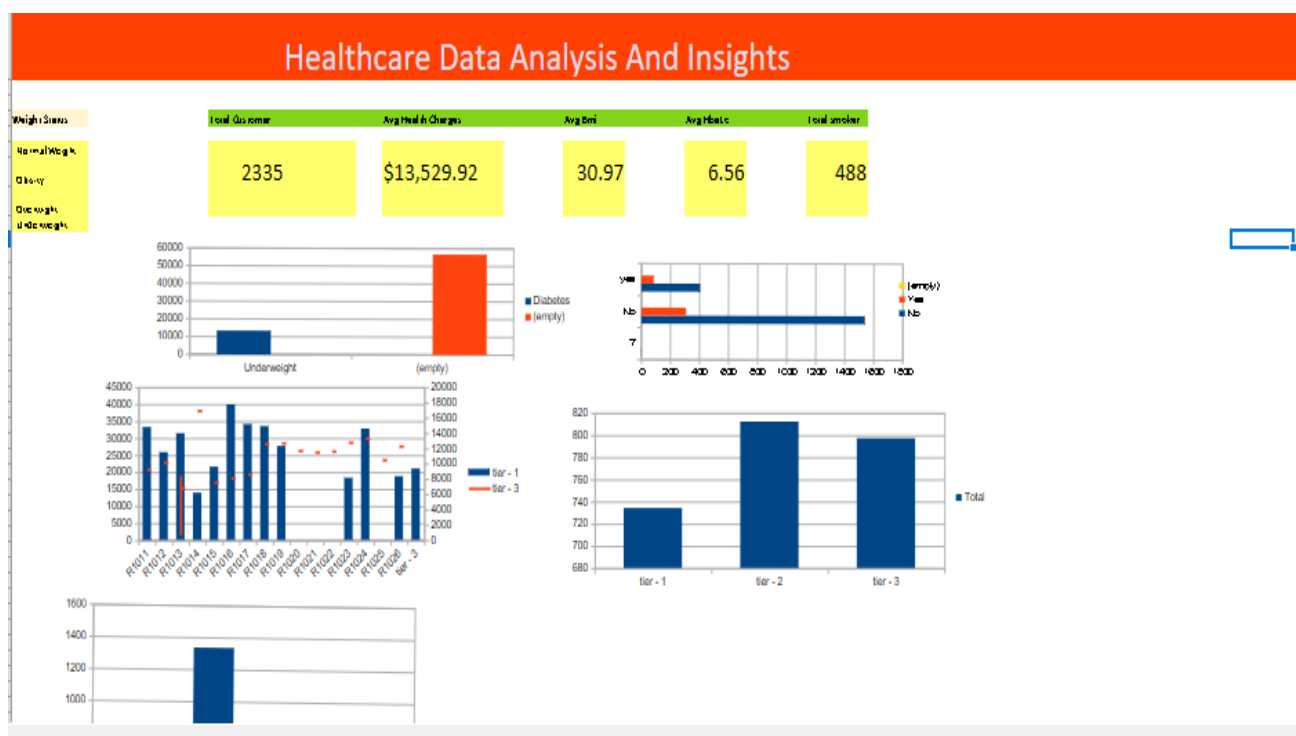
Pivot Table Analysis

O	P	Q	R	S	T	U	V	W	X	Y	Z
						Hospital tier					
Cancer history						tier - 1	tier - 2	tier - 3	tier - 3	Total Result	
No	Yes	(empty)	Total Result				309	1339	694	1	2343
2			2								
1538		307	1845								
404		84	488								
1944		391	2335								
						Average - charges					
						State ID	tier - 1	tier - 2	tier - 3	tier - 3	Total Result
City tier	Count - City tier*					R1011	₹33,181.29	₹18,997.28	₹9,250.36		₹19,557.6
tier - 1	734					R1012	₹25,800.40	₹10,312.05	₹10,179.94		₹11,990.3
tier - 2	812					R1013	₹31,371.85	₹8,459.98	₹6,857.11	₹700.00	₹10,538.4
tier - 3	797					R1014	₹13,891.86	₹2,395.17	₹16,950.49		₹13,478.0
Total Result	2343					R1015	₹21,523.52	₹16,433.91	₹7,570.76		₹14,489.6
						R1016	₹39,868.62	₹16,075.96	₹8,133.01		₹13,589.2
						R1017	₹34,070.60	₹11,365.62	₹8,667.37		₹14,806.1
						R1018	₹33,475.82	₹10,139.55	₹12,570.97		₹13,272.7
						R1019	₹27,621.61	₹8,116.72	₹12,706.98		₹13,633.1
						R1020		₹6,760.54	₹11,716.45		₹9,238.5
						R1021		₹7,668.18	₹11,518.79		₹8,933.3
						R1022		₹10,102.03	₹11,637.81		₹10,650.5
						R1023	₹18,261.76	₹13,457.09	₹12,785.64		₹13,786.1
						R1024	₹32,732.19	₹10,174.35	₹13,360.98		₹12,865.1
						R1025		₹6,886.04	₹10,510.31		₹8,788.7
						R1026	₹18,709.80	₹11,919.61	₹12,296.78		₹12,489.5
						tier - 3	₹20,996.53				₹20,996.5
Total Result							₹30,132.42	₹11,873.63	₹9,450.27	₹700.00	₹13,559.0
Hospital tier	Count - Hospital tier*										
tier - 1	309										
tier - 2	1339										
tier - 3	694										
tier - 3	1										
Total Result	2343										
smoker	Count - smoker*										
?	2										
No	1845										

Healthcare Master Sheet

	M	N	O	P	Q	R	S
1	charges	Hospital tier	City tier	State ID	Age		
2	₹563.84	tier - 2	tier - 3	R1013	30		
3	₹570.62	tier - 2	tier - 1	R1013	30		
4	₹600.00	tier - 2	tier - 1	R1013	29		
5	₹604.54	tier - 3	tier - 3	R1013	30		
6	₹637.26	tier - 3	tier - 3	R1013	24		
7	₹646.14	tier - 3	tier - 3	R1012	21		
8	₹650.00	tier - 3	tier - 3	R1013	30		
9	₹650.00	tier - 3	tier - 3	R1013	27		
10	₹668.00	tier - 3	tier - 2	R1012	20		
11	₹670.00	tier - 3	tier - 3	R1013	25		
12	₹687.54	tier - 3	tier - 2	R1013	21		
13	₹700.00	tier - 3	tier - 3	R1013	23		
14	₹722.99	tier - 3	tier - 1	R1013	23		
15	₹750.00	tier - 3	tier - 1	R1012	20		
16	₹760.00	tier - 3	tier - 1	R1013	29		
17	₹760.00	tier - 3	tier - 3	R1013	26		
18	₹770.00	tier - 3	tier - 3	R1013	29		
19	₹770.38	tier - 3	tier - 3	R1012	26		
20	₹773.54	tier - 3	tier - 2	R1013	27		
21	₹830.52	tier - 3	tier - 2	R1011	18		
22	₹865.41	tier - 3	tier - 1	R1013	22		
23	₹896.21	tier - 3	tier - 1	R1013	29		
24	₹915.07	tier - 3	tier - 1	R1013	28		
25	₹928.59	tier - 3	tier - 1	R1013	27		
26	₹964.71	tier - 3	tier - 2	R1013	21		

Healthcare Dashboard



Conclusion

The Healthcare Analytics Dashboard successfully transformed raw healthcare data into meaningful insights. This project demonstrates practical skills in Excel/LibreOffice including data cleaning, pivot tables, charts, KPI cards, dashboard creation, and healthcare analysis.